

# AI 504 Problem set 2

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Due Friday, March 13

**Instructions.** These are identical to the last problem set. Summary: you are allowed/encouraged to submit in groups; all students responsible for all problems; no adaptive online help.

**Grading.** A grade of 7 or higher means that you have *at understood all of the questions*. Even in the case that you do not completely solve a problem, write down your most coherent and intelligent response. On top of which, a 10 means a correct answer to at least three questions, and a 12 means a correct answer to all questions. Grades of 0, 2, or 4 are awarded on an ad-hoc basis.

1. Work over the set  $\{a, b, c, d\}$  of nouns. Consider the set of sentences

$$\Gamma = \{\text{All } a \text{ are } b, \text{ All } a \text{ are } c, \text{ All } b \text{ are } d, \text{ All } c \text{ are } d\}.$$

Write down **two different proof trees** verifying  $\Gamma \vdash \text{All } a \text{ are } d$ . Write your proof trees carefully and as legibly as possible.

2. Work over the set  $\{a, b, c, d\}$  of nouns. In this question I want you to find **two different models of the same size**. Each of your models should satisfy all of the following sentences

All  $a$  are  $b$ , All  $a$  are  $c$ , All  $b$  are  $d$ , All  $c$  are  $d$

and falsify both of the following sentences

All  $b$  are  $c$ , All  $c$  are  $b$ .

*Note.* We have to be a little careful in saying what it means for two models to be *different*. But basically if you label the elements of your model  $1, 2, \dots, n$  you should not be able to relabel (i.e., permute) the elements of one of your models to get the other one.

3. The *dual* of a sentence is obtained by switching its two nouns. For example the dual of All  $p$  are  $q$  is All  $q$  are  $p$ . We denote the dual of a sentence  $\varphi$  by  $\varphi^\dagger$ . If  $\Gamma$  is a set of sentences, by  $\Gamma^\dagger$  we mean the obvious thing, i.e., “take the dual of each sentence in  $\Gamma$ .”

Suppose  $\Gamma \vdash \varphi$ . Prove that  $\Gamma^\dagger \vdash \varphi^\dagger$  by induction on proof trees. Be very careful and deliberate, so that I know you understand what you’re doing.

4. Continuing the previous problem, there is also a notion of a *dual model*. So the idea is that given a model  $\mathcal{M}$ , we should be able to define its “dual”  $\mathcal{M}^\dagger$ , which has the property that  $\mathcal{M}^\dagger \models \varphi^\dagger$  if and only if  $\mathcal{M} \models \varphi$ , for each sentence  $\varphi$ . Define the dual and prove that it enjoys this property.
- (*Hint.* Start out with some explicit small models, see if you can find their duals by trial and error, and then see what the common pattern is.)

5. The point of this exercise is to show you that models can be built out of anything, as long as they have the correct type. Work over the set  $N = \{0, 1, 2, 3\}$  of nouns. Now this set of nouns makes for weird-looking sentences like All 3 are 1. But once we construct an  $N$ -model, it makes perfect sense to say that such sentences are satisfied or falsified.

Define the  $N$ -model  $\mathcal{R}$  as follows. Its domain is  $\mathbb{R}$ , the set of real numbers, and for each  $b \in N$ , define

$$[[b]] = \{x \in \mathbb{R} : x^b - x < 0\}.$$

List all of the  $N$ -sentences that  $\mathcal{R}$  satisfies, and argue that your answer is correct. (For the purposes of this exercise, assume  $0^0 = 1$ .)